

Ben M. Andrew

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Education

- University of Manchester**, PhD in Computer Science Sep 2024 – Present
- Research focused on formal methods for understanding degraded systems.
 - Supervised by [Dr. Marie Farrell](#), [Dr. Louise Dennis](#), and [Prof. Michael Fisher](#).
- University of Cambridge**, BA (Hons) in Computer Science Oct 2019 – Jun 2022
- Grade II.1 (70%).
 - Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by [Dr. Jackson Woodruff](#).

Publications

- Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M. *Counterexample-Guided Interval Weakening*. Rigorous State-Based Methods (ABZ) 2026. [10.1007/978-3-032-26752-8_1](https://doi.org/10.1007/978-3-032-26752-8_1)
- Andrew, B. M. *Weakening Goals in Logical Specifications*. Rigorous State-Based Methods (ABZ) 2025. [10.1007/978-3-031-94533-5_22](https://doi.org/10.1007/978-3-031-94533-5_22)

Industry Experience

- Software Engineer**, Tarides – Paris, France Sep 2022 – Jun 2024
- Developed open-source continuous integration (CI) software for the OCaml language’s ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
 - Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.
- Embedded Systems Intern**, Tarides – Paris, France Jul – Sep 2021
- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.
- Computer Graphics Research Intern**, University of Cambridge Jun – Sep 2020
- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Professional Activities

- Part of the local organising committee of Integrated Formal Methods (iFM) 2024.
- Member of the [Autonomy and Verification Network](#).

Teaching Experience

- University of Cambridge**, Supervisor (TA equiv.) Oct 2022 – Present
- Leading small group supervisions for undergraduate courses in computer science, including topics such as logic, formal verification, and semantics of programming languages.
- University of Manchester**, Graduate Teaching Assistant Oct 2024 – May 2026
- Assisting in teaching graduate and undergraduate courses in computer science, including topics such as software engineering, logic, and formal verification.